Attention GANs: Unsupervised Deep Feature Learning for Aerial Scene Classification

Yunlong Yu, Xianzhi Li, *Student Member, IEEE*, and Fuxian Liu

***Abstract*— With the development of deep learning, supervised feature learning methods have achieved prominent performance in the field of aerial scene classification. However, supervised feature learning methods require a large amount of labeled training data. To address this limitation, in this article, a novel unsupervised deep feature learning method, namely, Attention generative adversarial networks (Attention GANs), is proposed for aerial scene classification. First, Attention GANs integrates the attention mechanism into GANs to enhance the representation power of the discriminator. Then, to obtain contextual information, a context-aggregation-based feature fusion architecture is designed in the discriminator. Furthermore, the generator and discriminator losses are improved on basis of the Relativistic GAN. At the same time, a content loss is formed by using the feature representations from the context-aggregation-based feature fusion architecture. In the experiments, our Attention GANs is evaluated via comprehensive experiments with four publicly available remote sensing scene data sets, i.e., the UC-Merced data set with 21 scene classes, the RSSCN7 data set with 7 scene classes, the AID data set with 30 scene classes, and the NWPU-RESISC45 data set with 45 scene classes. Experimental results demonstrate that our Attention GANs can obtain the best performance compared with the state-of-the-art methods.**

***Index Terms*— Aerial scene classification, attention mechanism, context aggregation, generative adversarial networks (GANs), unsupervised deep feature learning.**

I. Introduction

A HUGE amount of remote sensing images are gradually accessible recently with the rapid development of the satellite imaging techniques and the growing deployment of the remote sensing instruments [1]–[4]. Analyzing and understanding these remote sensing images, as for identification and classification, have become more and more urgent for both civil and military purposes [5]–[11]. In particular, aerial scene classification, which aims to automatically label an aerial scene image based on a set of semantic categories, has been a hot topic. However, due to the large intraclass variations and the small interclass dissimilarity, extracting robust and discriminative representations on aerial scene images still poses an unsolved problem.

Besides traditional methods using handcrafted features [12]–[15], deep feature learning-based approaches are widely explored to address the bias introduced by handcrafted features. Among them, convolutional neural network (CNN)-based approaches [16]–[20], as a crucial branch of supervised deep feature learning based methods, have yielded amazing performance. However, these methods require a large amount of labeled images for training, which is difficult to satisfy in most application scenarios due to the annotation efforts.

In light of these limitations, there is growing interest in learning features in an unsupervised manner. In particular, the generative adversarial networks (GANs) have been shown to excel in many applications such as image classification [21], segmentation [22], and super-resolution [23], to name a few. Very recently, Lin *et al.* [24] first proposed to employ GANs on the aerial scene classification task. As the primary attempt, however, there is still much room for further improvement. In the process of training GANs, whether the sample image is real or fake is known. The real and fake do not represent the label of the image in this article. In this article, the label of one image represents the semantic category, e.g., airport, church, desert, and so on. All unsupervised feature learning methods do not require any labeled data in the process of feature learning, in which the label all represents the semantic category.

In this article, we present a novel framework, namely *Attention GANs*, to first conduct feature learning in an unsupervised setting, and then classify aerial scene images via a simple support vector machines (SVMs). To design such a workable GANs framework to solve the classification task, first, we enhance the representation power of the discriminator by integrating the attention mechanism into GANs. Note that the idea of attention-based GANs has been explored in several works, e.g., Self-Attention GANs [25] and Attention-Aware GANs [26]; however, they aim to solve the image generation tasks. To better fit our aerial scene classification task, we deliberately employ max-pooling, L_p-pooling and average-pooling to squeeze the spatial information and the channel information of the input feature map. Second, we design a context aggregation-based feature fusion architecture in the discriminator, in which the dilated convolutional layer is employed to obtain contextual information. Furthermore, we improve the generator loss and the discriminator loss based on Relativistic GAN. At the same time, the feature

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representations from the context aggregation-based feature fusion architecture are utilized to form a content loss. Once the Attention GANs is trained, the discriminator network can be regarded as the feature extractor. The extracted feature representations are fed into the SVMs for classification. We extensively evaluate the effectiveness of our Attention GANs on various aspects, as well as on four widely used benchmarks, i.e., the UC-Merced data set, the RSSCN7 data set, the AID data set, and the NWPU-RESISC45 data set. Experimental results demonstrate that our proposed Attention GANs obtains the best performance when compared with the state-of-the-art unsupervised feature learning methods.

Overall, the contributions of this work are summarized as follows.

1) We develop a novel unsupervised framework, i.e., Attention GANs, to conduct feature representation learning for aerial scene classification. To the best of our knowledge, no prior works attempt to solve the aerial scene classification task by integrating the attention mechanism into GANs.
2) We design a new context aggregation-based feature fusion architecture in the discriminator network and adopt dilated convolutional layer to get contextual information.
3) We employ the idea of Relativistic GAN to improve the generator loss and discriminator loss. Furthermore, we propose to use the feature representations after the context aggregation-based feature fusion architecture to form a content loss.

The rest of this article is arranged as follows. In Section II, the related works are briefly reviewed. Section III gives a brief introduction of the preliminary knowledge including GANs and deep convolutional GANs (DCGANs). Section IV presents the detailed description of our proposed framework. In Section V, we show the experimental results on four widely used data sets. Finally, Section VI summarizes this article and presents our ideas for the future work.

II. Related Works

Recently, the topic of aerial scene classification has been widely explored due to its extensive applications. With respect to the types of features used for aerial scene classification, the existing methods can be broadly categorized into four groups: handcrafted feature-based methods, unsupervised shallow feature learning-based methods, supervised deep feature learning-based methods, and unsupervised deep feature learning-based methods. In the following, we briefly review each category.

A. Handcrafted Feature-Based Methods

With the handcrafted feature-based methods, it is supposed that the main characteristic of an aerial scene image can be represented by some handcrafted features, i.e., texture, color, spectral, and structure information. The most widely used methods include scale-invariant feature transform (SIFT) [12], local binary pattern (LBP) [13], color histogram (CH) [14], and GIST [15]. It is worth noting that these handcrafted feature-based methods perform well on some aerial scene images with homogeneous spatial arrangements. However, these approaches are difficult to depict the abundant semantic information in aerial scenes, which would lead to the bad performance in practical applications.

B. Unsupervised Shallow Feature Learning-Based Methods

In order to improve the performance of handcrafted feature-based methods, many research works have developed unsupervised feature learning-based methods for aerial scene classification, in which features can be automatically learned from the unlabeled scene images. In this section, we only introduce the unsupervised shallow feature learning-based methods. The aim of unsupervised shallow feature learning-based methods is to form a set of basis filters which are used for feature encoding. The input of the filters is usually a set of handcrafted features, e.g., SIFT, CH, and LBP of local image patches, and the output is a holistic representation of the input scene image. Typical unsupervised shallow feature learning approaches include sparse coding [6], [27]–[33], k-means clustering [34]–[40], principal component analysis [41]–[43], and autoencoder [44]–[46]. Compared with handcrafted feature based methods, unsupervised shallow feature learning-based methods can get more abstract and discriminative features. However, these methods still do not have sufficient ability to deal with the aerial scene images with more complicated structures and spatial layouts.

C. Supervised Deep Feature Learning-Based Methods

In more recent years, CNN, a crucial branch of the methods of deep learning, has been attracting more research attention due to its ability of automatically discovering informative and significant features on image classification tasks. Almost all of the supervised deep feature learning-based methods employ CNN as their network backbone. Those CNNs are trained in a supervised way via backpropagation, which have been obtained remarkable performance on aerial scene classification tasks. There is a vast literature on the supervised deep feature learning-based methods [16]–[20], [47]–[54], which can be categorized into the following three groups. The general idea of the first group [17], [47]–[50], [54] is to employ full-trained, fine-tuned, or pretrained CNN to extract features from the input scene images, and then feed these features into the classifier such as SVM, extreme learning machine (ELM), and K-nearest neighbor (KNN) for classification. The methods in the second group [16], [18]–[20], [51] generate the final scene image representation by encoding the features extracted from the pretrained or fine-tuned CNN. In the last group [52], [53], they adopt some fusion techniques, i.e., feature-level fusion, and score-level fusion, to serve the process of aerial scene classification. The methods in the last group all use pretrained or fine-tuned CNN model.

Despite the success in aerial scene classification, these approaches *all* need labels for training. In contrast, our method solves the feature learning task in an unsupervised manner.

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D. Unsupervised Deep Feature Learning-Based Methods

Unsupervised deep feature learning-based methods have been a paramount aspect of remote sensing, since they can extract more powerful features from unlabeled data compared to the unsupervised shallow feature learning-based methods. Stacked autoencoder (SAE) [55], as an early and relatively simple deep learning framework for aerial scene classification, is composed of multiple layers of sparse autoencoders. In SAE, the outputs of each layer are connected to the inputs of the successive layer. However, 1-D vector of the input in SAE can result in the damage of the spatial information derived from the aerial scene images.

As a pioneer work, Romero *et al.* [56] proposed an unsupervised deep CNN for feature learning, where the model was trained in a greedy layerwise manner. However, it is not an end-to-end architecture. Tao *et al.* [57] presented an end-to-end learning system named unsupervised-restricted deconvolutional neural network (URDNN) by employing the stacked convolutional autoencoder to conduct parameter initialization. Mou *et al.* [58] also introduced an end-to-end network called deep Conv–Deconv neural network, which is constructed by the convolutional subnetwork and deconvolutional subnetwork. Recently, the multiple-layer feature-matching GANs (MARTA GANs) [24] was proposed by first introducing GANs to solve the aerial scene classification task.

III. Preliminaries

For better understanding of our proposed GANs based framework, here, we briefly review some preliminary works.

A. GANs

GANs [59] consist of a generator G and a discriminator D that compete in a two-player minimax game. The discriminator tries to distinguish real training data from synthetic data, and the generator tries to produce fake samples and fool the discriminator; see Fig. 1 as an illustration.

The concept of GANs has been applied to many computer vision tasks, e.g., image super-resolution [60], [61], image generation [62], unsupervised feature learning [24], and semisupervised feature learning [63]. Although GANs has led to significant improvements in unsupervised learning, it suffers from the unstable training process and often results in generators that produce unreasonable outputs.

B. DCGAN

To address this limitation and further boost the performance of GANs, DCGANs [64] was proposed by integrating CNNs into both generator and discriminator, but with the following modifications. First, DCGAN replaces any pooling layers with deconvolutional layers and strided convolutions for generator and discriminator, respectively. Second, DCGAN employs batch normalization and removes all the fully connected layers to stabilize the learning. Finally, DCGAN uses rectified linear unit (ReLU) activation function in generator for all layers except for the output, which uses Tanh function instead. Meanwhile, it uses LeakyReLU in the discriminator for all layers.

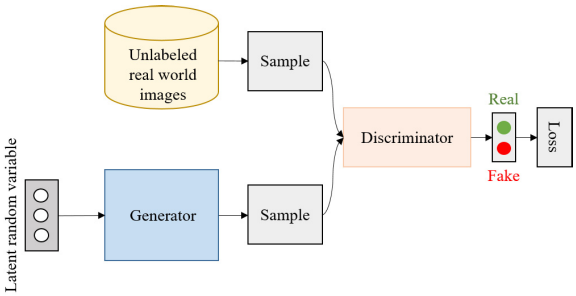


Fig. 1. Typical architecture of GANs.

IV. Method

A. Overview

Fig. 2 shows the overall architecture of our proposed Attention GANs. The generator network [see Fig. 2(a)] produces fake images from some random numbers drawn from a uniform distribution. The discriminator network [see Fig. 2(b)], as the adversary of the generator, tries to distinguish between real images and fake images. Since the discriminator serves as the feature extractor, to further boost the performance, we introduce the attention module into it; see the yellow arrows in Fig. 2(b). Meanwhile, we also design a novel context aggregation based feature fusion; see the right part of Fig. 2(b). In the following, we first present the attention module in Section IV-B. Next, we show how to conduct context aggregation based feature fusion in Section IV-C. Finally, we introduce the improved generator loss and discriminator loss in Section IV-D.

B. Attention Module

Attention mechanism plays a significant role in human visual system, in which some salient areas are selected to focus after conducting a sequence of partial glimpses [65]. Various approaches have been proposed to integrate attention mechanism into CNNs [66]–[68]. The good performance of attention mechanism in these works indicates that it can not only present what and where are informative but can also further improve the feature representation ability. Inspired by this, we design the novel attention module and plug it into the discriminator [see the yellow arrows in Fig. 2(b)] to enhance the feature extraction ability. The design of our attention module borrows ideas from SENet [68], but makes some improvements.

Specifically, the proposed attention module used in our framework is divided into two submodules, i.e., the channel attention submodule, and the spatial attention submodule, which are shown in Fig. 3. These two attention submodules are connected in a sequential manner. The intermediate feature map $\mathbf{F} \in \mathbb{R}^{C\times H\times W}$ is first fed into the channel attention submodule, and then the channel-refined feature map $\mathbf{F_c} \in \mathbb{R}^{C\times H\times W}$ is taken as the input of the spatial attention submodule. The whole procedure of the attention module is as follows:

$$\mathbf{F_c} = \mathbf{O_c} \otimes \mathbf{F} \tag{1}$$

$$\mathbf{F_{cs}} = \mathbf{O_s} \otimes \mathbf{F_c} \tag{2}$$

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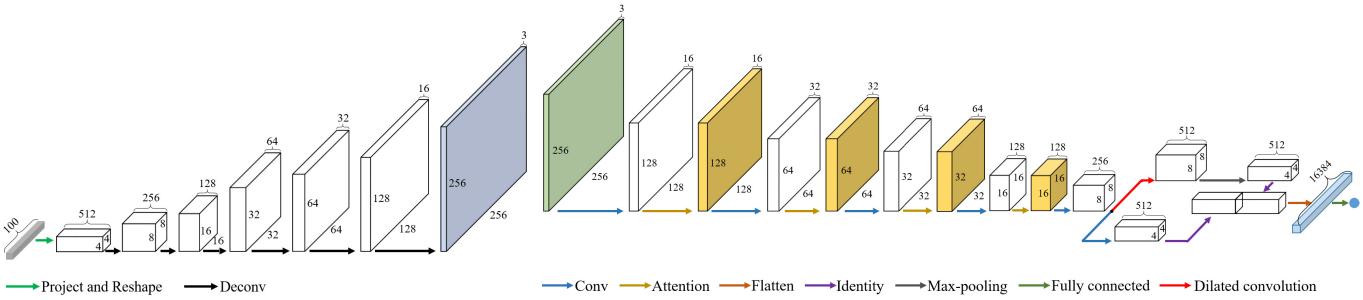


Fig. 2. Overall architecture of the proposed Attention GANs. (a) Generator of Attention GANs. (b) Discriminator of Attention GANs.

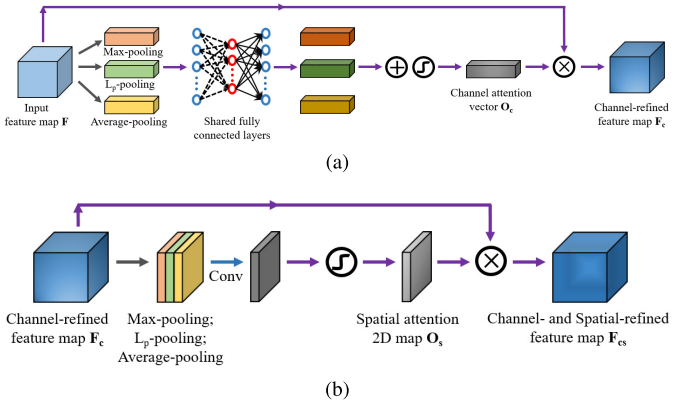


Fig. 3. Architecture of the proposed attention module used in our Attention GANs. (a) Channel attention submodule. (b) Spatial attention submodule.

where $\mathbf{O_c}$ represents the channel attention vector, $\mathbf{O_s}$ denotes the spatial attention 2-D map, $\mathbf{F_{cs}}$ is the channel- and spatial-refined feature map, and $\otimes$ represents elementwise multiplication. The following part gives the details of these two submodules.

1) Channel Attention Submodule: In order to aggregate abundant information in each channel, we employ max-pooling, L_p-pooling [69] and average-pooling to squeeze the spatial information of the input feature map, respectively. With regard to squeeze spatial dimension, the max-pooling and average-pooling operations have been widely employed. In addition, L_p-pooling can be seen as a tradeoff between the max-pooling and average-pooling, which could gather other distinctive clues. Therefore, we use max-pooling, L_p-pooling and average-pooling simultaneously. After the squeezing operation, we can get three channel vectors, where the size of each channel vector is $\mathbb{R}^{C\times 1\times 1}$. Then, each of these three channel vectors is fed into the shared fully connected layers (SFCLs) with one hidden layer, in which the number of hidden units U is less than the input dimension C. After employing elementwise summation and activation to the three channel vectors, we can obtain the channel attention vector $O_c \in \mathbb{R}^{C\times 1\times 1}$. Thus, the channel-refined feature map $\mathbf{F_c}$ is computed as follows:

$$\begin{aligned}\mathbf{F_c} = \sigma\big(&\mathrm{SFCL}(\mathrm{MaxPool}_s(\mathbf{F})) \\ &+ \mathrm{SFCL}(L_p\mathrm{Pool}_s(\mathbf{F})) \\ &+ \mathrm{SFCL}(\mathrm{AvgPool}_s(\mathbf{F}))\big) \otimes \mathbf{F} \end{aligned} \quad (3)$$

$$\mathrm{SFCL}(\mathrm{MaxPool}_s(\mathbf{F})) = W_2(\delta(W_1(\mathrm{MaxPool}_s(\mathbf{F})) + b_1)) + b_2 \quad (4)$$

$$\mathrm{SFCL}(L_p\mathrm{Pool}_s(\mathbf{F})) = W_2(\delta(W_1(L_p\mathrm{Pool}_s(\mathbf{F})) + b_1)) + b_2 \quad (5)$$

$$\mathrm{SFCL}(\mathrm{AvgPool}_s(\mathbf{F})) = W_2(\delta(W_1(\mathrm{AvgPool}_s(\mathbf{F})) + b_1)) + b_2 \quad (6)$$

where δ refers to the ReLU function, σ represents the sigmoid function, $W_1 \in \mathbb{R}^{U\times C}$ and $W_2 \in \mathbb{R}^{C\times U}$ are the weights in SFCL, and $b_1 \in \mathbb{R}^U$ and $b_2 \in \mathbb{R}^C$ are the biases in SFCL.

2) Spatial Attention Submodule: Different from channel attention submodule focusing on what is significant, the spatial attention submodule pays attention to where is the important part. Thus, these two submodules are complementary parts. In order to achieve better result for spatial attention, we also employ the three pooling operations used in channel attention submodule, i.e., max-pooling, L_p-pooling, and average-pooling, to squeeze the channel information of the input feature map. These three operations are applied along the channel dimension. Then, we concatenate these three 2-D maps and put them into the convolution layer with a sigmoid activation function. After the activation function, we can obtain the spatial attention 2-D map $\mathbf{O_s} \in \mathbb{R}^{1\times H\times W}$, which can show where should be highlighted or restrained. Finally, the channel- and spatial-refined feature map $\mathbf{F_{cs}}$ is computed as follows:

$$\mathbf{F_{cs}} = \sigma\left(\mathrm{conv}([\mathrm{MaxPool}_c(\mathbf{F}); L_p\mathrm{Pool}_c(\mathbf{F}); \mathrm{AvgPool}_c(\mathbf{F})])\right) \otimes \mathbf{F_c} \quad (7)$$

where σ represents the sigmoid activation function, $[\cdot;\cdot;\cdot]$ represents the concatenation operation and $conv$ denotes the convolution operation.

C. Context Aggregation-Based Feature Fusion

Feature fusion has become an effective and robust way to obtain rich feature representations. In our discriminator, we propose the context aggregation-based feature fusion strategy by combining the features extracted via the dilated convolutional layer [70] and the conventional convolutional layers, respectively; see the right part of Fig. 2(b).

In detail, we first employ the dilated convolutional layer to get contextual information; see the red arrow in Fig. 2(b). Suppose that m represents the input feature map and w denotes

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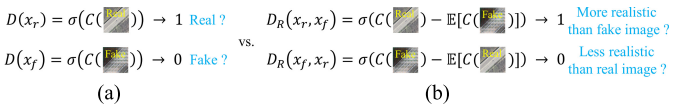


Fig. 4. Difference between the traditional discriminator and the relativistic discriminator. (a) GANs. (b) Relativistic GANs.

the weights of the dilated convolution operator, the output y of the dilated convolution operator is defined as follows:

$$y(p) = \sum_{d \in K_d} w(d) \cdot m(p + d) \tag{8}$$

where p represents the position, K_d denotes the grid for the dilated filters, and d is the dilation factor.

The features extracted by dilated convolutional layer are fused with the features extracted by conventional convolutional layer. In other words, the designed architecture contains two streams, i.e., one stream with conventional convolution operator and the other with dilated convolution operator. With regard to feature fusion strategy, we adopt a simple but effective way, i.e., serial feature fusion strategy, to concatenate two sets of features into one set of features. By using the proposed context aggregation-based feature fusion architecture, we can obtain the visual details of the area around each pixel of the input feature map as well as its larger context.

D. Loss Functions

Besides the attention module and the context aggregation-based feature fusion architecture, we also propose to improve the loss functions used in our framework. We first employ the idea of Relativistic GAN [71] to make the discriminator predict relative real probability instead of an absolute one. Then, we use the feature representations after the context aggregation based feature fusion architecture to form a content loss.

Different from the traditional discriminator D, which emits the probability value that the input image is a real sample, the relativistic discriminator D_{R} estimates the probability that the real image is more realistic than the fake one. Fig. 4 shows the difference between the traditional discriminator and the relativistic discriminator. Note that we refer to real image as x_r and fake image as x_f, $C(x)$ denotes the nontransformed discriminator output, σ represents the sigmoid function and $\mathbb{E}[\cdot]$ is the operation of taking average of all specified data in the minibatch. Then, the $D_{\mathrm{R}}(x_r - x_f)$ and $D_{\mathrm{R}}(x_f - x_r)$ are computed as follows:

$$D_{\mathrm{R}}(x_r - x_f) = \sigma(C(x_r) - \mathbb{E}_{x_f \sim \mathbb{Q}}[C(x_f)]) \tag{9}$$

$$D_{\mathrm{R}}(x_f - x_r) = \sigma(C(x_f) - \mathbb{E}_{x_r \sim \mathbb{P}}[C(x_r)]) \tag{10}$$

where $\mathbb{Q}$ represents the distribution of the fake data and $\mathbb{P}$ denotes the distribution of the real data. Thus, the discriminator loss L_D^R is formulated as

$$L_D^R = -\mathbb{E}_{x_r \sim \mathbb{P}}[\log(D_{\mathrm{R}}(x_r, x_f))] - \mathbb{E}_{x_f \sim \mathbb{Q}}[\log(1 - D_{\mathrm{R}}(x_f - x_r))]. \tag{11}$$

Next, we can get the adversarial loss of the generator L_G^R as follows:

$$L_G^R = -\mathbb{E}_{x_r \sim \mathbb{P}}[\log(1 - D_{\mathrm{R}}(x_r, x_f))] - \mathbb{E}_{x_f \sim \mathbb{Q}}[\log(D_{\mathrm{R}}(x_f, x_r))]. \tag{12}$$

We can note that the adversarial loss of the generator L_G^R is affected by x_r and x_f simultaneously, while, in traditional GANs, only x_f is considered. Thus, this modification can help our generator network get benefits from the gradients from both real data and fake data.

In order to make the generator network draw more realistic images, we consider the Euclidean distance between the context aggregation-based feature fusion representations of the real image and the fake image. Let F' denotes the feature representations after the operation of concatenating, the content loss L_{content} for our generator network is defined as follows:

$$L_{\mathrm{content}} = \|\mathbb{E}_{x_r \sim \mathbb{P}} F'(x_r) - \mathbb{E}_{x_f \sim \mathbb{Q}} F'(x_f)\|_2 \tag{13}$$

where $\|\cdot\|_2$ represents the L_2 norm. Finally, the total loss of the generator L_G is

$$L_G = L_{\mathrm{content}} + L_G^R. \tag{14}$$

V. Experiments

In this section, we first introduce the data sets used for the experiments and the implementation details of our network. Then, we perform extensive ablation study to thoroughly evaluate the effectiveness of our design choice. Finally, we give the confusion analysis and the state-of-the-art comparisons, respectively. The code and models will be publicly available.[1]

A. Data Set Description

1) UC-Merced Data Set: The first commonly used data set is the UC-Merced data set [72], consisting of 2100 overhead scene images with 256 × 256 pixels evenly from 21 classes in the red, green, and blue (RGB) color space. These images are selected from the United States Geological Survey (USGS) National Map; see Fig. 5 for the examples, where each image is randomly selected from each category.

2) RSSCN7 Data Set: The second data set is called RSSCN7 data set [73], which is collected from Google Earth. It contains 2800 aerial scene images with 400 × 400 pixels from 7 classes. There are 400 images in each category. Fig. 6 shows the examples which are randomly selected from each class.

3) AID Data Set: The third data set, AID data set [48], is also collected from Google Earth, which contains 10 000 images from 30 classes with 600 × 600 pixels. The class size varies from 220 to 420; see Table I for the details. Fig. 7 shows the examples from each class. Note that images in this data set are captured from different countries in different seasons.

[1] https://github.com/yunlong-yu/Attention-GANs

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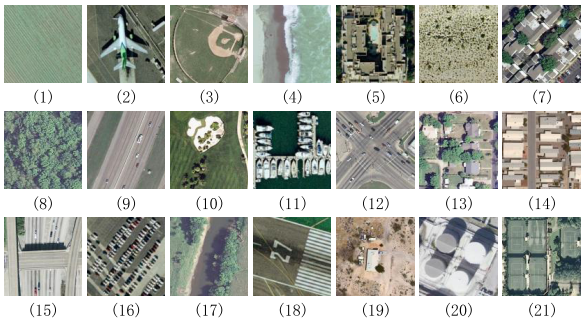


Fig. 5. Example images in the UC-Merced data set: (1) agricultural, (2) airplane, (3) baseball diamond, (4) beach, (5) buildings, (6) chaparral, (7) dense residential, (8) forest, (9) freeway, (10) golf course, (11) harbor, (12) intersection, (13) medium residential, (14) mobile home park, (15) overpass, (16) parking lot, (17) river, (18) runway, (19) sparse residential, (20) storage tanks, and (21) tennis court.

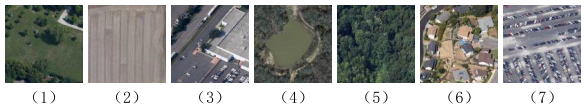


Fig. 6. Example images in the RSSCN7 data set: (1) grass, (2) field, (3) industry, (4) river lake, (5) forest, (6) resident, and (7) parking.



Fig. 7. Example images in the AID data set: (1) airport, (2) bare land, (3) baseball field, (4) beach, (5) bridge, (6) center, (7) church, (8) commercial, (9) dense residential, (10) desert, (11) farmland, (12) forest, (13) industrial, (14) meadow, (15) medium residential, (16) mountain, (17) park, (18) parking, (19) playground, (20) pond, (21) port, (22) railway station, (23) resort, (24) river, (25) school, (26) sparse residential, (27) square, (28) stadium, (29) storage tanks, and (30) viaduct.

4) NWPU-RESISC45 Data Set: The fourth data set, NWPU-RESISC45 [49], is a large-scale remote sensing scene data set, which contains 31 500 images with 256×256 pixels evenly from 45 scene classes. The spatial resolution varies from about 30–0.2 m per pixel. The characteristics of this data set include large scale, high within class diversity, high between class similarity, and rich image variations.

TABLE I
NUMBER OF SCENE IMAGES IN EACH CATEGORY OF THE AID DATA SET

Name	♯images	Name	♯images	Name	♯images
airport	360	farmland	370	port	380
bare land	310	forest	250	railway station	260
baseball field	220	industrial	390	resort	290
beach	400	meadow	280	river	410
bridge	360	medium residential	290	school	300
center	260	mountain	340	sparse residential	300
church	240	park	350	square	330
commercial	350	parking	390	stadium	290
dense residential	410	playground	370	storage tanks	360
desert	300	pond	420	viaduct	420

B. Implementation Details

In our experiments, we use a workstation equipped with the Intel Xeon(R) E5-1620 processor with 10 M of Cache and up to 3.6 GHz, 16-GB memory, an NVIDIA GeForce GTX1080Ti GPU, and a SAMSUNG SSD 850 PRO with 1 TB of capacity. Our method is implemented using TensorFlow.

In the generator network, the ReLU activation function is used in all deconvolutional layers except for the output layer, where we use the tanh function instead. In the discriminator network, the LeakyReLU activation function with a slope of 0.2 is used in all convolutional layers except for the output layer, where we use the sigmoid function instead. The kernel size of all the convolutional and deconvolutional layers is set to 4×4 and the stride is set to 2. We use the training samples with a mini-batch size of 64 and train our networks using the Adam optimizer with a learning rate of 0.0002.

To evaluate the effectiveness of feature representation of our Attention GANs comprehensively, we adopt two different training-testing ratios for each utilized data set. For the UC-Merced data set, the ratio of training/testing images is set to 50:50 and 80:20, respectively. For the RSSCN7 data set and the AID data set, we fix the ratio of the training set to be 20% and 50%, respectively, and the remaining images for testing. For the NWPU-RESISC45 data set, the ratio of training/testing images is set to 10:90 and 20:80, respectively. We use the same training ratios when training our Attention GANs and the SVMs in all data sets, which is consistent with other unsupervised methods. We compute the two commonly used metrics including the overall accuracy and the confusion matrix to compare the performance quantitatively. Furthermore, in order to obtain reliable results, the final results are reported as the mean and standard deviation over ten runs.

Note that the training process of our Attention GANs does not require any labeled data. Once the Attention GANs is trained, the discriminator network will be regarded as an effective feature extractor. The extracted features are fed into the SVMs for classification.

C. Ablation Study

In this part, we empirically analyze the major elements in our framework by removing or replacing each of them when we train our framework: 1) *Without Attention*—we remove the attention module from the discriminator network; 2) *Without Dilation*—we remove the stream with dilated convolution

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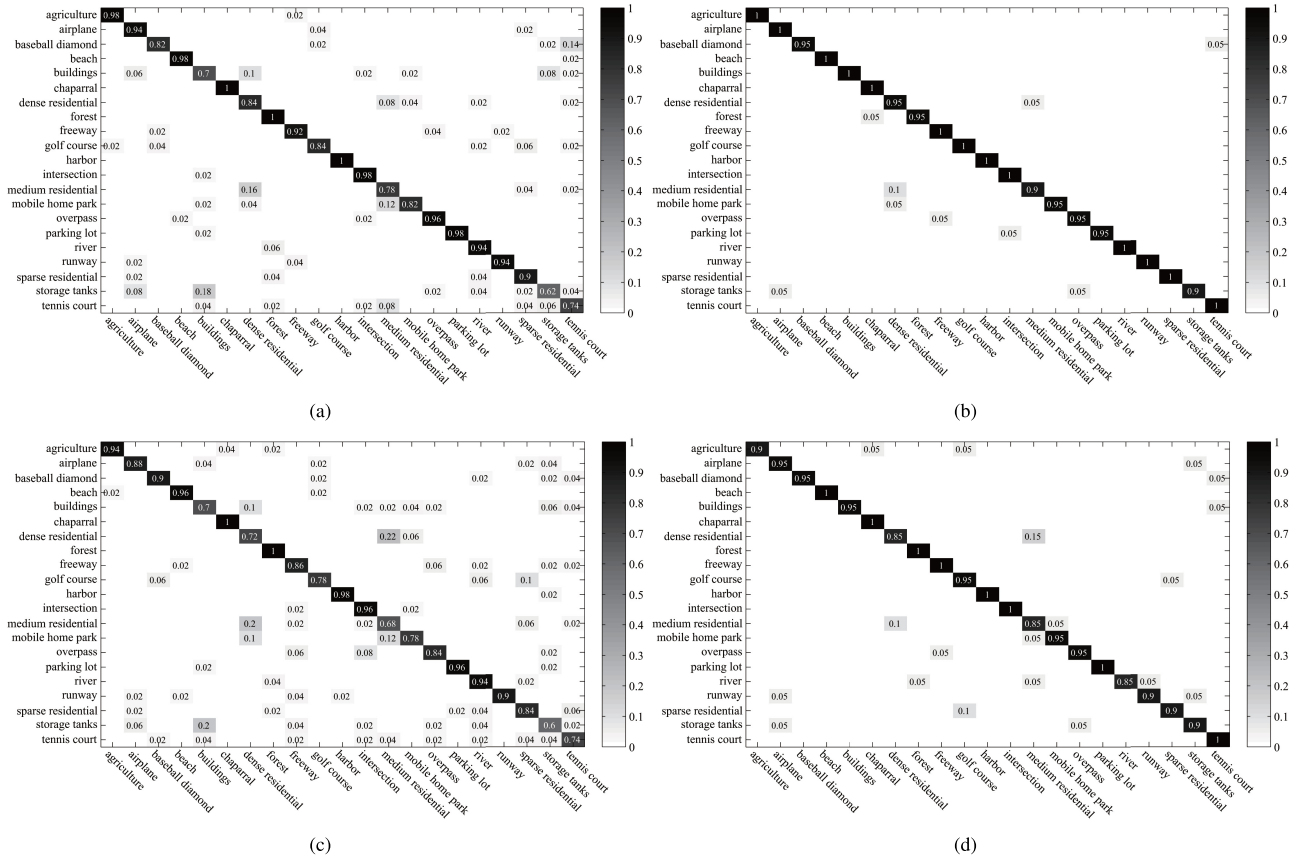


Fig. 8. Confusion matrixes of the UC-Merced data set under the training ratios of 50% and 80% using two different methods. (a) Attention GANs on the 50% training ratio. (b) Attention GANs on the 80% training ratio. (c) MARTA GANs on the 50% training ratio. (d) MARTA GANs on the 80% training ratio.

TABLE II

COMPARING THE CLASSIFICATION ACCURACY BY EMPLOYING DIFFERENT SETTINGS ON THE UC-MERCED DATA SET

Different settings	Training ratios (%)	
	50%	80%
Baseline	83.90 ± 0.56	92.20 ± 0.71
w/o Attention	86.60 ± 0.60	95.95 ± 0.81
w/o Dilation	86.76 ± 0.58	95.98 ± 0.77
Traditional Loss	88.21 ± 0.52	96.90 ± 0.72
Our full pipeline	89.06 ± 0.50	97.69 ± 0.69

TABLE III

COMPARING THE CLASSIFICATION ACCURACY BY EMPLOYING DIFFERENT SETTINGS ON THE RSSCN7 DATA SET

Different settings	Training ratios (%)	
	20%	50%
Baseline	77.28 ± 0.60	82.86 ± 0.57
w/o Attention	81.03 ± 0.68	85.70 ± 0.58
w/o Dilation	81.15 ± 0.68	85.99 ± 0.57
Traditional Loss	82.18 ± 0.76	86.42 ± 0.55
Our full pipeline	83.47 ± 0.63	87.32 ± 0.54

operator; 3) *Traditional Loss*—we replace relativistic loss with the traditional loss for unsupervised learning; and 4) *Baseline*—we remove the attention module and the stream with dilated convolution operator, and, at the same time, we replace relativistic loss with the traditional loss.

Tables II–IV show the overall accuracies and standard deviations of employing four different settings on the UC-Merced, RSSCN7, and AID data sets, respectively. From the results in Tables II–IV, we can observe the following.

1) The classification accuracies of our full pipeline achieve 89.06%, 97.69%, 83.47%, 87.32%, 78.95%, and 84.52% on the UC-Merced data set (50% and 80% training samples), the RSSCN7 data set (20% and 50% training samples), and the AID data set (20% and 50% training samples), respectively, which outperform all the other settings. The good performance of our full pipeline reveals that the framework elements being explored here all contribute to produce significant feature representations and improve the classification accuracy.
2) Comparing the results produced with the *Traditional Loss* setting and with the *Without Attention* setting, we can see that the accuracy reduction (compared with the classification accuracy of our full pipeline) of the *Without Attention* setting is higher than that of the *Traditional Loss* setting, thus indicating that the contribution of our attention module is more remarkable than the relativistic loss. Looking at the results produced with the

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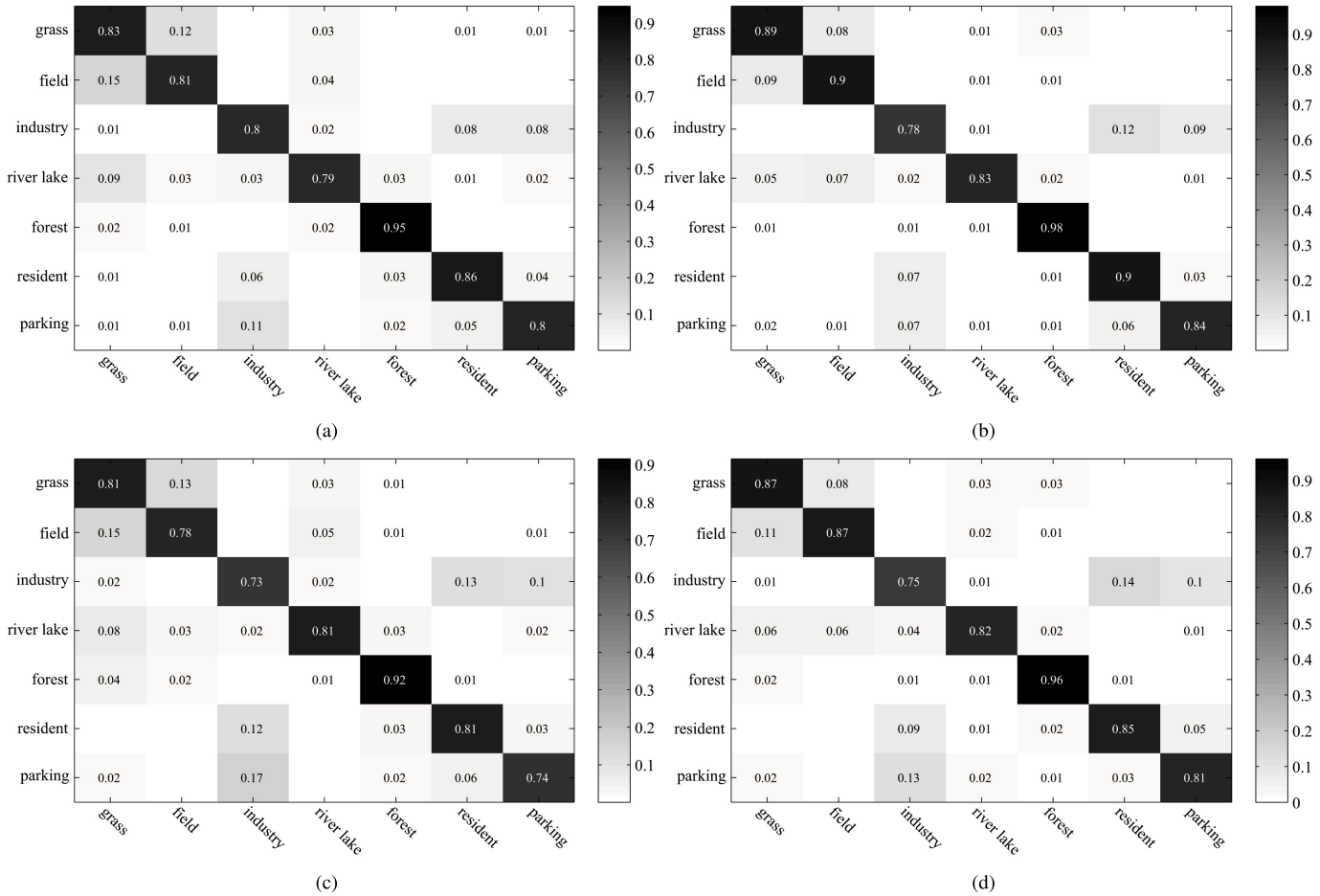


Fig. 9. Confusion matrixes of the RSSCN7 data set under the training ratios of 20% and 50% using two different methods. (a) Attention GANs on the 20% training ratio. (b) Attention GANs on the 50% training ratio. (c) MARTA GANs on the 20% training ratio. (d) MARTA GANs on the 50% training ratio.

TABLE IV
COMPARING THE CLASSIFICATION ACCURACY BY EMPLOYING DIFFERENT SETTINGS ON THE AID DATA SET

Different settings	Training ratios (%)	
	20%	50%
Baseline	73.59 ± 0.38	79.95 ± 0.17
w/o Attention	76.55 ± 0.37	82.89 ± 0.19
w/o Dilation	76.67 ± 0.34	82.91 ± 0.16
Traditional Loss	77.95 ± 0.33	83.77 ± 0.19
Our full pipeline	78.95 ± 0.23	84.52 ± 0.18

Without Attention and *Without Dilation* settings, we can see that both of them get almost the same accuracy reduction compared with the classification accuracy of our full pipeline, which illustrates that the importance of our attention module and the stream with dilated convolution operator are almost equal.

D. Confusion Analysis

In order to further validate the performance of our proposed framework, we give the results measured in terms of the confusion matrix. Due to the limitation of the article space, we only compare the confusion matrices of our Attention GANs with that of MARTA GANs, which is a pioneer work to employ GANs to conduct unsupervised feature learning for aerial scene classification. In the confusion matrix, the value in the *i*th row and *j*th column represents the rate of test images from the *i*th category that are predicted as the *j*th class. Figs. 8–10 show the confusion matrixes on the UC-Merced data set, RSSCN7 data set, and AID data set, respectively. The followings can be observed from Figs. 8–10.

1) From the confusion matrices on the UC-Merced data set, we can see that our Attention GANs performs better than MARTA GANs. Under the 80% training samples, we find that the confusion errors obviously decrease. For instance, the confusion errors of agriculture/chaparral, agriculture/golf course, airplane/storage tanks, buildings/tennis court, golf course/sparse residential, medium residential/mobile home park, mobile home park/medium residential, river/forest, sparse residential/golf course reduce by 100%. The confusion error of dense residential/medium residential reduces by 66.67%. Furthermore, there are 13 categories with classification accuracies of 1 in the confusion matrix produced by our Attention GANs, while MARTA GANs only get eight classes with classification accuracies of 1. Under the 50% training samples, Attention GANs obtains better

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TABLE V
COMPARISON OF CLASSIFICATION ACCURACY WITH THE STATE-OF-THE-ART METHODS ON THE UC-MERCED DATA SET

Method	Training ratios (%)	
	50%	80%
LLC [48]	70.12 ± 1.09	72.55 ± 1.83
BoVW [48]	72.40 ± 1.30	75.52 ± 2.13
BoVW+SCK [72]	-	77.80
Dense Sift [29]	-	81.67 ± 1.23
With-sal [46]	-	82.72 ± 1.18
IFK [48]	78.74 ± 1.65	83.02 ± 2.19
CNN-1 [56]	-	84.53
Li's method [40]	-	89.10
UFL-SC [74]	-	90.26 ± 1.51
SCMF [28]	-	91.03 ± 0.48
Q-OMP [32]	-	92.29 ± 0.71
MARTA GANs [24]	85.51 ± 0.69	94.86 ± 0.80
Lu's method [33]	-	95.71
Ours	**89.06 ± 0.50**	**97.69 ± 0.69**

TABLE VI
COMPARISON OF CLASSIFICATION ACCURACY WITH THE STATE-OF-THE-ART METHODS ON THE RSSCN7 DATA SET

Method	Training ratios (%)	
	20%	50%
LLC [48]	73.29 ± 0.63	77.11 ± 1.29
BoVW [48]	76.91 ± 0.59	81.28 ± 1.19
MARTA GANs [24]	79.86 ± 0.93	84.49 ± 0.62
Ours	**83.47 ± 0.63**	**87.32 ± 0.54**

TABLE VII
COMPARISON OF CLASSIFICATION ACCURACY WITH THE STATE-OF-THE-ART METHODS ON THE AID DATA SET

Method	Training ratios (%)	
	20%	50%
LLC [48]	58.06 ± 0.50	63.24 ± 0.44
BoVW [48]	62.49 ± 0.53	68.37 ± 0.40
MARTA GANs [24]	75.39 ± 0.49	81.57 ± 0.33
Ours	**78.95 ± 0.23**	**84.52 ± 0.18**

TABLE VIII
COMPARISON OF CLASSIFICATION ACCURACY WITH THE STATE-OF-THE-ART METHODS ON THE NWPU-RESISC45 DATA SET

Method	Training ratios (%)	
	10%	20%
BoVW+SPM [49]	27.83 ± 0.61	32.96 ± 0.47
LLC [49]	38.81 ± 0.23	40.03 ± 0.34
BoVW [49]	41.72 ± 0.21	44.97 ± 0.28
MARTA GANs [24]	68.63 ± 0.22	75.03 ± 0.28
Ours	**72.21 ± 0.21**	**77.99 ± 0.19**

or at least the same per-class accuracies for 20 out of 21 classes compared to MARTA GANs.

2) From the confusion matrices on the RSSCN7 data set, we can observe similar phenomenon that our Attention GANs can provide consistent improvement in the classification performance on most scene classes compared to MARTA GANs. Meanwhile, the misclassifications can be significantly reduced. For example, Attention GANs can get higher per-class accuracies for six out of seven categories, seven out of seven categories under the 20% and the 50% training samples, respectively.
3) When analyzing the confusion matrices on the AID data set, similarly, our Attention GANs gives better performance than MARTA GANs. Our Attention GANs can get relatively good performance under some scene types which are difficult to distinguish. We take the training ratio of 50% for example. From the matrix obtained by using MARTA GANs, we can observe the scene classes with low accuracy, i.e., industrial (0.69), resort (0.62), school (0.55), and square (0.60). Our Attention GANs can improve the classification accuracy rates of these four scene classes, i.e., industrial (0.70), resort (0.66), school (0.64), and square (0.62).

E. State-of-the-Art Comparison

Next, we compare the performance of our method against several state-of-the-arts quantitatively. To be fair, these state-of-the-arts are all unsupervised feature learning methods. Tables V–VIII show the comparison results on UC-Merced, RSSCN7, AID, and the NWPU-RESISC45 data sets, respectively. Note that the best values are marked in bold. From these tables, we can see that, overall, our method could achieve the highest classification accuracies on all data sets compared with others.

1) On the UC-Merced data set, under the 50% training ratio, our method has 3.55% higher classification accuracy in comparison with MARTA GANs, which is the second-best method; see the third row from the bottom versus the bottom row in Table V. Besides, under the 80% training ratio, the classification accuracy of our method is still 1.98% higher compared with the second-best method, i.e., Lu's method [33]; see the second row from the bottom versus the bottom row in Table V.
2) On the RSSCN7 data set, compared with the second-best method, i.e., MARTA GANs (see the third row in Table VI), our method can achieve an additional 3.61% and 2.83% improvement using 20% and 50% training ratios, respectively; see the bottom row in Table VI.
3) On the AID data set, our method gains the overall accuracy improvements with 3.56% and 2.95% over the second-best method, i.e., MARTA GANs, under the training ratios of 20% and 50%, respectively.

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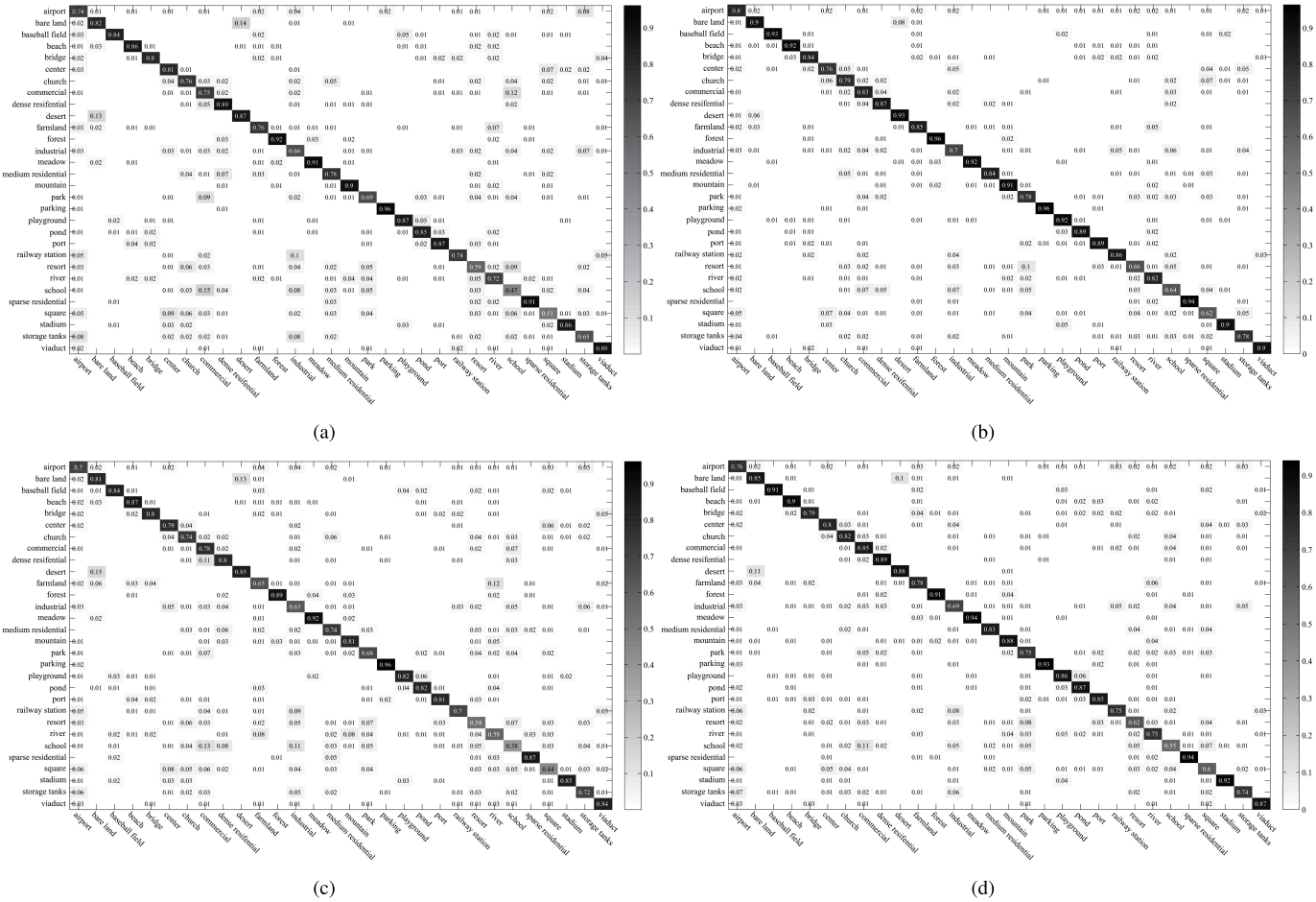


Fig. 10. Confusion matrixes of the AID data set under the training ratios of 20% and 50% using two different methods. (a) Attention GANs on the 20% training ratio. (b) Attention GANs on the 50% training ratio. (c) MARTA GANs on the 20% training ratio. (d) MARTA GANs on the 50% training ratio.

4) On the NWPU-RESISC45 data set, our method obtains the overall accuracy improvements with 3.58% and 2.96% over the second-best method, i.e., MARTA GANs, under the training ratios of 10% and 20%, respectively.

The good performance of our Attention GANs benefits from using GANs enhanced by the improved attention module, the context aggregation-based feature fusion architecture, and the modified loss functions.

F. Comparison With Pretrained and Fine-Tuned CNN-Based Methods

In addition, we also compare the performance of our Attention GANs against some pretrained and fine-tuned CNN-based methods on the relatively big data sets, i.e., AID and NWPU-RESISC45 data sets. In the pretrained CNN-based methods, all CNN models are pretrained on the ImageNet [75] in a supervised manner. Then, the pretrained CNNs are used for feature extraction. With regard to the fine-tuned CNN-based methods, the pretrained CNN models are fine-tuned on the target aerial scene data sets in a supervised manner. Next, the fine-tuned architectures are served as feature extractor. In this section, we do not employ the implementation details used in previous experiments.

TABLE IX
COMPARISON OF CLASSIFICATION ACCURACY WITH PRETRAINED AND FINE-TUNED CNN-BASED METHODS ON THE AID DATA SET

Method	Training ratios (%)	
	20%	50%
Pre-trained CNN-based methods		
VGG-VD-16 [48]	86.59 ± 0.29	89.64 ± 0.36
CaffeNet [48]	86.86 ± 0.47	89.53 ± 0.31
DCA with concatenation [76]	-	89.71 ± 0.33
Fusion by concatenation [76]	-	91.86 ± 0.28
Fusion by addition [76]	-	91.87 ± 0.36
TEX-Net-LF [52]	90.87 ± 0.11	92.96 ± 0.18
VGG-VD16+MSCP [77]	91.52 ± 0.21	94.42 ± 0.17
Ours (Pre-trained)	92.30 ± 0.24	94.93 ± 0.21
Fine-tuned CNN-based methods		
Multilevel fusion (Fine-tuned) [53]	-	95.36 ± 0.22
Ours (Fine-tuned)	93.97 ± 0.23	96.03 ± 0.16

For fair comparison, in the following experiments, we use ImageNet [75] to pretrain our Attention GANs in an unsupervised fashion. Then, the pretrained model is fine-tuned on the target aerial scene data sets in an unsupervised manner, which is used for the comparison with other fine-tuned CNN-based methods.

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TABLE X
COMPARISON OF CLASSIFICATION ACCURACY WITH PRETRAINED AND FINE-TUNED CNN-BASED METHODS ON THE NWPU-RESISC45 DATA SET

Method	Training ratios (%)	
	10%	20%
Pre-trained CNN-based methods		
AlexNet (Pre-trained) [49]	76.69 ± 0.21	79.85 ± 0.13
LASC-CNN [17]	81.37	84.30
BoCF (VGGNet-16) [51]	82.65 ± 0.31	84.32 ± 0.17
VGG-VD16+MSCP [77]	85.33 ± 0.17	88.93 ± 0.14
Ours (Pre-trained)	86.11 ± 0.22	89.44 ± 0.18
Fine-tuned CNN-based methods		
VGGNet-16 (Fine-tuned) [49]	87.15 ± 0.45	90.36 ± 0.18
Ours (Fine-tuned)	88.06 ± 0.19	90.58 ± 0.23

Tables IX and X show the comparison results on AID and the NWPU-RESISC45 data sets, respectively. From these results, we can see that our method can achieve better classification accuracy than other methods. At the same time, our method has the advantage of small network size. The innovation of the architecture and the loss functions makes our method perform well.

VI. CONCLUSION

In this article, a novel unsupervised deep feature learning method, Attention GANs, is proposed for aerial scene classification. First, Attention GANs integrates the attention mechanism into the GANs. Then, a context aggregation-based feature fusion architecture is designed to obtain contextual information. Furthermore, the generator loss and the discriminator loss are improved on basis of the Relativistic GAN. At the same time, the feature representations from the context aggregation-based feature fusion architecture are utilized to form a content loss. In the end, extensive experiments are conducted on four publicly available remote sensing scene data sets. According to the experimental results, our Attention GANs achieves the promising performance compared with state-of-the-art methods on all four data sets.

Inspired by our the promising results produced from our Attention GANs, in the future, we plan to extend our framework to other applications, e.g., hyper-spectral image classification and multi-spectral image classification, *etc.*. Besides, we may also explore to further improve our generator network to yield more realistic images, since the current one is relatively simple only with a series of deconvolutional layers.

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